

L_RAG Presentation Slide Modifications Report

Text Mining Final Project Presentation Updates

September 18, 2026

Summary of Modifications

This document summarizes the exact updates applied to the presentation deck `Text_Mining.pdf` during the current prompt session. All changes align with the official ablation report (`ablation_report.pdf`) and incorporate feedback regarding Knowledge Graph schema construction, multi-stage pipeline breakdown, and cross-attention reranking insights.

Slide #	Action	Summary of Change
Slide 6	INSERTED (NEW)	L_RAG Pipeline: Stage-by-Stage Processing: Breaks down the 4 runtime stages (Indexing, KG Overlay Joiner, RRF Fusion & Rerank, Grounded LLM Gen).
Slide 7	INSERTED (NEW)	G-LRAG Knowledge Graph: Schema & Construction: Details node hierarchy (Docs, Provisions, Chunks), edge types, and ingested scale (151k docs, 1.38M provisions, 1.51M chunks).
Slide 14	MODIFIED (UP-DATED)	Key Reranker Insight: Replaced placeholder " <i>Additional diagnostics</i> " box with technical cross-attention interaction insight.

Table 1: Summary table of slide deck modifications.

Detailed Breakdown of New Slides

1. Slide 6: Stage-by-Stage Pipeline Processing (Inserted after Slide 5)

- **Stage 1 (Indexing):** Metadata enrichment (Title || Header || Passage) + underthesea BM25 term tokenization.
- **Stage 2 (KG Overlay):** Dynamic `id_str_filter` validity masking + 1-hop `CHUNK_NEXT` reading-order expansion.
- **Stage 3 (Fusion & Rerank):** RRF ($k = 60$) candidate fusion ($K = 30$) + GPU load-run-unload Cross-Encoder reranking ($N = 10$).
- **Stage 4 (Generation):** `gpt-4o-mini` ($T = 0.0$) with statutory citation grounding and `INSUFFICIENT_CONTEXT` abstention.

2. Slide 7: Knowledge Graph Schema and Construction (Inserted after Slide 6)

- **Graph Node Hierarchy:** Formalizes Document (D), Provision (P), and Chunk (C) entity nodes.
- **Edge Schema:** Distinguishes Structural edges (DOC_HAS_PROVISION, PROVISION_HAS_CHUNK, CHUNK_NEXT, PROVISION_NEXT) from Legal Relations (amend, replace, cite, repeal).
- **Ingested Scale Table:** Embeds the formal corpus statistics table (151,624 Documents, 1,386,267 Provisions, 1,513,376 Chunks, and 883,256 Document Edges).
- **Validity Rule:** Highlights dynamic as_of_date whitelisting and quarantining of unverified records.

Detail of Slide Update (Slide 14)

Replaced the low-priority placeholder "*Additional diagnostics*" box with the core technical justification:

"Rerankers jointly score each query–candidate pair; capturing richer cross-attention interactions than comparing separately encoded query and document vectors."

Verification and Compilation

The updated slide source file docs/slide/Text_Mining.tex was recompiled cleanly using xelatex. The total slide count expanded from 19 to 21 slides without modifying any existing subsequent slides.